

# ETHAN PARK

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## EDUCATION

### University of Waterloo

*Bachelor of Applied Science, Nanotechnology Engineering*

- Recipient of University of Waterloo President's Scholarship

**Sep 2023 - Apr 2028**

Waterloo, ON

## EXPERIENCE

### Cell Engineering Intern

*National Research Council Canada*

- Delivered 40x increase in Li-ion cathode slurry throughput by programming a liquid-dispensing robot and 3D printing custom labware for seamless integration into a self-driving lab (SDL).
- Assembled and characterized 2000+ Li-ion half cells using CV, EIS, and OCP, and accelerated data analysis by 36x using Python to automate data visualization, noise reduction, and peak finding.
- Investigated NMC phase compositions (XRD), oxygen-assisted PCAM calcination, and Niobium/Tungsten doping concentrations using Bayesian optimization to achieve specific capacities within 90% of literature values.

**Sep 2025 - Apr 2026**

Mississauga, ON

### Research Assistant, Redox Flow Battery Design

*Van Der Heijden Lab*

- Improved mass transport across electrode interfaces by 50% using SolidWorks CFD to optimize flow field and inlet designs.
- Reduced electrolyte leakage and mechanical warping by 90% in 3D printed flow cells through iterative optimization of SLA printing parameters, resin selection, and curing processes.
- Investigated nitrogen-assisted carbonization conditions to convert highly porous polymer structures to functional electrodes.

**Apr 2025 - Aug 2025**

Waterloo, ON

### Electrical Engineering Intern

*Smith + Andersen*

- Drafted electrical device and power distribution drawings in AutoCAD for 15+ public infrastructure projects.
- Completed electrical load and lighting calculations to support energy-efficient layouts, reducing client expenses by 20%.
- Conducted engineering site audits to monitor project progress and incorporated P.Eng. review feedback to validate CAD deliverables for contractors.

**Jan 2025 - Apr 2025**

Toronto, ON

## EXTRACURRICULAR

### Semiconductor Process Development Team

*Waterloo Hacker Fab*

- Designed a test bench for I-V characterization of fabricated semiconductor devices to support electrical testing.
- Achieved 3 micron patterning resolution through spin coating and lithography process development.
- Investigating thermal oxidation parameters using a custom tube furnace to optimize silicon oxide growth for microfabrication processes and improve process repeatability.

**May 2026 - Present**

Waterloo, ON

### Class Representative

*Waterloo Engineering Society*

- Elected to represent the Nanotechnology cohort, serving as a key liaison between students and society executives
- Gained strong communication and presentation skills by contributing budgetary solutions during society discussions
- Engaged hundreds of engineering students across campus by collaborating with fellow society reps to organize and promote large-scale student social events

**Sep 2023 - Jan 2025**

Waterloo, ON

## TECHNICAL SKILLS

**Characterization Techniques:** XRD, SEM, EIS, CV, OCP, UV-Vis, IV

**Battery Processing:** NMC Cathode Synthesis, Slurry Compounding, Electrode Fabrication, Cell Assembly

**Micro Fabrication:** Photolithography, Spin Coating, Thin Film Growth, Thermal Oxidation, Cleanroom ISO 4

**Programming:** Python (NumPy, SciPy, Pandas), Matlab, Git

**CAD and Simulation:** SolidWorks, COMSOL, AutoCAD, Simulink